Week 11: 9/11 – 13/11

Subject: Math - Statistics **Unit Title: Unit 2: Describing and Displaying Data** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: The Five-Number Summary and Boxplots	• I can determine the five-number summary and construct and interpret a boxplot.	1.2 Alternatives for auditory information (captions, transcripts, visuals). 8.3 Foster collaboration and community with structured group roles and supports. Choose an item.	Links to support learning: Chebyshev's Theorem Standard Deviation Formula, Statistics, Variance, Sample and Population Mean
Lesson 2: Outliers and Identifying Outliers Using IQR	• I can use the interquartile range to identify and interpret outliers in a dataset.		
Lesson 3: Summative Assessment:	• I can calculate and interpret population variance and standard deviation using the correct notation. • I can apply the Chebyshev's and Empirical rule • I can calculate and interpret z scores to describe and compare the relative positions of values within their datasets.		
Lesson 4: Boxplots and Outliers: Application	• I can construct and interpret a boxplot and identify outliers in a real-world dataset.		
Homework	• Complete any assigned Homework in your Notebook	8.1 Make goals clear, visible, and supported by short-term objectives. Choose an item.	
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments.		

Technology Integration	Microsoft (Power point, excel, etc.)
Cross-curricular links	**English / Media Literacy:** Students interpret survey questions, headlines, and data-based claims, then communicate and justify conclusions using clear evidence.
External Assessment Integration	SAT
Wellbeing/ ATL Skills	**Thinking Skills – Critical Thinking:** Students analyse data sources, identify bias, classify information, and evaluate whether conclusions are supported by the data. **Communication Skills:** Students discuss findings, justify choices, and use appropriate statistical vocabulary. **Research Skills:** Students plan data collection and evaluate the reliability and relevance of information.
Standards	• **HSS-ID.A.1 -** Represent numerical data using plots on the real number line, specifically boxplots. • **HSS-ID.A.3 -** Interpret distribution shape, center, and spread in context, considering extreme values and outliers.
Inquiry/ TOK Question	**How can we collect and evaluate data to make reliable conclusions about a population?**
Success Criteria (Measurable)	**Lesson 1** • **Must:** I must be able to identify the minimum, first quartile, median, third quartile, and maximum of a dataset. • **Should:** I should be able to construct an accurately scaled boxplot using the five-number summary. • **Could:** I could interpret the median, box, and whiskers to explain the spread and possible skewness of a distribution. **Lesson 2** • **Must:** I must be able to calculate the interquartile range by subtracting the first quartile from the third quartile. • **Should:** I should be able to identify outliers by calculating the lower and upper fences using one point five times the interquartile range. • **Could:** I could justify whether a value is an outlier using the calculated fences and interpret its significance in context. **Lesson 3: Summative Assessment** **Lesson 4** • **Must**: I must be able to determine the five-number summary and construct an accurately scaled boxplot. • **Should:** I should be able to calculate the interquartile range and use the lower and upper fences to identify outliers. • **Could:** I could use the boxplot and identified outliers to explain the distribution's spread and possible skewness in context.

National Identity Domain	Volunteering التطوع	Global Understanding التفاهم العالمي	Heritage التراث
	Area: Community Responsibility and Informed Decision-Making		
	Implementation: Using annual rainfall data from a UAE location, students determine the five-number summary, construct a boxplot, and apply the IQR rule to identify unusually wet or dry years. Students explain how understanding rainfall variability can support water conservation and responsible management of the UAE's natural resources.		